Deep Freezer



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 5 labs

Covers these lab items: Deep Freezer; Freezers; Vertical four door Freezer

What it is used for

A deep freezer keeps food and other approved materials at low temperatures to maintain quality and slow spoilage. Correct temperature control, hygienic loading, and regular monitoring help protect stored items and ensure reliable operation.

Main parts

- Insulated cabinet and door maintain the cold storage environment.
- Compressor circulates refrigerant through the cooling system.
- Evaporator removes heat from the freezer compartment.
- Condenser releases heat from the refrigerant.
- Thermostat or digital controller regulates the set temperature.
- Shelves, baskets, or racks organise stored items.
- Door gasket and latch help maintain an airtight seal.
- Temperature display and alarm indicate operating conditions on equipped models.

Before you start

1. Read the user manual and identify the required storage temperature for the stored materials.
2. Check that the freezer is level, ventilated, and positioned away from heat sources.
3. Inspect the power cord, plug, door gasket, shelves, and cabinet for damage.
4. Confirm the interior is clean, dry, and free from unwanted odours.
5. Verify the temperature display and alarm function where available.
6. Prepare labelled, sealed containers and ensure items are suitable for freezer storage.

How to use

1. Connect the freezer to a suitable power supply according to the manual.
2. Switch on the freezer and set the recommended temperature.
3. Allow the compartment to reach the required temperature before loading items.
4. Arrange labelled containers with space for air circulation and avoid blocking vents.
5. Close the door firmly and check that the gasket seals properly.
6. Record the temperature at the required monitoring intervals.
7. Minimise door opening and return items promptly after each retrieval.
8. Remove frost using the approved defrosting procedure when necessary.
9. If cleaning or defrosting requires shutdown, transfer stored items to suitable temperature-controlled storage first.

10. Clean and dry the compartment, restore power if required, and verify the temperature before reloading.

⚠ Safety

- Never touch electrical plugs or controls with wet hands.
- Do not use the freezer if the cord, plug, or cabinet wiring is damaged.
- Keep ventilation openings clear and do not obstruct the condenser area.
- Never chip ice with sharp tools or use open flames to defrost the freezer.
- Store food in sealed, labelled containers and separate incompatible materials.
- Follow safe lifting practices when loading heavy containers or shelves.

Care & storage

- Clean interior surfaces and removable shelves using approved cleaning materials.
- Inspect the door gasket regularly and clean it to maintain a proper seal.
- Keep the condenser and ventilation areas clear of dust according to the manual.
- Arrange authorised servicing if cooling performance, alarms, or electrical operation becomes abnormal.

Common mistakes to avoid

- Loading warm items before the freezer reaches its required temperature.
- Overfilling shelves and blocking internal air circulation.
- Leaving the door open or ignoring temperature alarms.
- Removing ice with sharp tools or unsuitable heating methods.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.